

PVR: Creating personalized cinematic experiences with a high- performance data warehouse powered by **PVR** BigQuery

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ABOUT PVR

About PVR

PVR is the largest and the most premium film exhibition company in India. Since its inception in 1997, the brand has redefined the way entertainment is perceived in the country. PVR currently operates a cinema circuit comprising of 860 screens at 179 properties in 73 cities (India and Sri Lanka), serving over 100 million patrons annually.

Industries: Media & Entertainment

To simplify its data analytics process, PVR provides personalized cinematic experiences. PVR leverages BigQuery as a centralized data

warehouse and g
data insights with
than 12X higher €

Location: India

Google Cloud (<https://cloud.google.com/>)

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BigQuery (<https://cloud.google.com/bigquery/>)

Google Cloud results:

- Synchronizes data collected from edge devices across 174 cinemas in a few minutes with BigQuery
- Enables a small team of five to manage data engineering and automation, freeing up more time to create personalized services
- Shortens time needed

Reduced
query time from
hours to
minutes

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Data Catalog

(<https://cloud.google.com/data-catalog>)

to generate
data
analytics
reports by up
to 12X

- Supports
data storage
and analytics
with 40%
higher cost
efficiency

Going to the cinema is no longer just about watching a newly released movie. As films are increasingly premiered online at the same time as hitting the big screen, more people now expect to enjoy an ultimate film-viewing experience at movie theaters. Beyond just excellent image and sound quality, audiences expect immersive 4D effects, and other premium services like special seats and varied food options.

PVR (<https://www.pvrcinemas.com/>) is an industry leader in India dedicated to delivering the best-possible cinematic experiences to different audience groups. Currently operating 846 screens in 176 cinemas across 71 cities in India and Sri Lanka, the company offers diverse content tailored for regional movie goers. It offers a wide range of cinema services such as child-friendly auditoriums, the latest screening technologies, in-seat food services, and even drive-in cinemas. It also provides apps for consumers to book tickets, food, taxis, receive coupons, and browse through its movie library.

“Data is very important for our business development because we want to create highly personalized products. We need powerful data storage and analytics tools that can scale easily and enhance our data processing efficiency while keeping the costs as low as possible. Google Cloud ticks all the boxes.”

—Ranjith Melarkode,
Strategic Technical
Officer, PVR

During the lockdown period of COVID-19 in 2020, the PVR team realized that it had to create more personalized cinema experiences by leveraging the power of data to attract cinema-goers and continue its business expansion once cinemas reopened. Back then, the company had an on-premises data center in each of its cinema stores and used a manual approach to aggregate all the data for analytics, which took just a few days to generate insights. At the same time, as people in India increasingly pick up digital habits since the onset of COVID-19, the amount of data that PVR collects from its apps started to grow exponentially, posing a huge challenge on the scaling capability of its on-premises data centers.

To build a data pipeline that supports higher data analytics efficiency and scalability, PVR decided to set up a centralized data warehouse. After comparing different public cloud offerings, it adopted Google Cloud (<https://cloud.google.com/>) in late 2020 for its streamlined data pipeline design, scalability, and high cost efficiency for data storage and analytics.

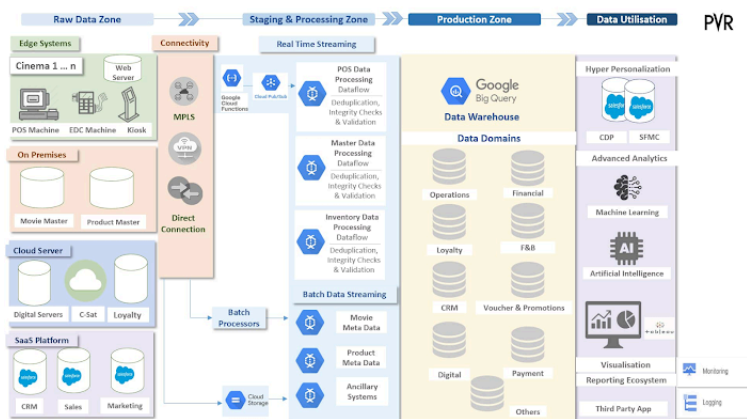
“Data is very important for our business development because we want to create highly personalized products,” says Ranjith Melarkode, Chief Information Officer at PVR. “We need powerful data storage and analytics tools that can scale easily and enhance our data processing efficiency while keeping the costs as low as possible. Google Cloud ticks all the boxes.”

A centralized data warehouse that supports swift synchronization

PVR now runs a data warehouse on BigQuery (/bigquery) that ingests streaming transaction data from edge devices like PoS machines, ticket QR code scanners and payment devices across all its 174 cinemas through Pub/Sub (/pubsub). The data warehouse also streams information from PVR's other on-premises and cloud systems such as customer service platforms, devices, digital apps, and loyalty management systems. Before the adoption of Google Cloud, the PVR team could only gather its data in one place at the end of every business day. With the new data pipeline, all data is synchronized every few minutes.

The new way of processing data also enables PVR to keep its data management team nimble. Previously, the company had around 10 employees monitoring data transfer and data accuracy. Now it only needs five personnel to manage the automated data pipelines, aggregation and reconciliation of numbers, reporting and data visualization, which has freed up the team to conduct more in-depth data analysis to deliver business insights needed to more personalized cinema services and products.

"The way our data is aggregated has been greatly optimized after we started using Google Cloud. Since we can receive the latest data across all our cinema stores almost in real time, we are able to make decisions related to movie performance, product (food) sales, stock, and inventory more promptly and effectively," explains Melarkode.



Reducing data query time for personalized product development

To facilitate the development of its personalized products, PVR built a customer data platform on a third-party digital marketing platform in February 2021. For the system to generate useful insights, the team needed to compile all its customer data collected over the past five years and transfer it from Google Cloud to the other platform. Melarkode points out that since BigQuery has helped significantly reduce the data query time, the team managed to complete the migration in under six weeks.

“We’re very impressed by how much BigQuery enabled us to accelerate the data migration process. In terms of data query, what used to take us 12 hours can now be done in several minutes. Without BigQuery, it would have taken us months to set up our customer data system,” he adds.

The customer data platform has since allowed the PVR team to offer more personalized services at an individual level by understanding its customers’

preferences better and tracking customer behavior across digital and physical storefronts. It can now send alerts to customers in real time on its apps about new offers that match their interests, or reminders on vouchers that are about to expire. For example, a customer walking into a cinema may receive offers on food with a personalized value voucher.

“Before migrating our data analytics workload to BigQuery, it used to take us around 12 hours to generate a business insight report. On BigQuery, a report can be produced in one hour, which has enabled us to conduct more analyses and

maximize our
productivity.”

—Ranjith Melarkode,
Strategic Technical
Officer, PVR

Generating data insights 12X faster with higher cost efficiency

Besides insights into customer preferences, PVR also employs BigQuery to generate business insights. These cover food revenue, costs and inventory, movie performance, people metrics (attendance, shifts), customer demographics, cinema performance (one store in retail parlance), along with analysis across groups of cinemas, regionally aggregated nationally. Previously, teams were heavily reliant on the Data team to ingest, clean, and validate numbers across different business groups, then publish figures for further analysis.

Melarkode adds that with the centralized data warehouse on BigQuery, the time required to produce data analytics reports has been greatly shortened, because each department can now conduct data analytics on the platform simultaneously, instead of queuing for analytics resources like before. This entire process is now done for multiple business teams with data available in near real time for a constant feedback loop of data and process refinement. On top of that, the PVR

team uses Data Catalog (/data-catalog) to discover, describe and organize data from all cinema sources. Further, Data catalog is also used to ensure that the data collected from all its cinema stores is consistent by tracking changes in metadata, which helps prevent errors and further enhance the data team's productivity.

"Before migrating our data analytics workload to BigQuery, it used to take us up to 12 hours to generate certain business insights. On BigQuery, similar reports and insights are produced in a few minutes, which has enabled us to conduct more analyses and maximize our productivity," says Melarkode.

Another major benefit that PVR enjoys from the migration to Google Cloud is the enhanced cost efficiency. Since the fees for storage and compute are decoupled on Google Cloud, PVR can store a large amount of data on the cloud platform and only pay for the computing resources used on demand for analytics, which normally involves only a small portion of the data. According to Melarkode, generating an insight report on Google Cloud is 40% more cost-efficient than any other cloud data analytics platforms.

"The more we
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**Maximizing the value of data with
artificial intelligence**

PVR is ready to continue its plan for business expansion to 1000+ screens and double-digit digital user growth YoY. Following the end of the lockdown measures in India, it plans to leverage the artificial intelligence (AI) capabilities of Vertex AI (/vertex-ai) to provide more accurate product recommendations based on its customers' purchase and engagement history. The predictive analytics powered by AI can also help the company better arrange its resources, such as preparing food according to demand predictions to ensure freshness and reduce waste.

Melarkode concludes, "The more we unlock the value of our data, the better services we can provide to our customers. With the high-performance data processing and analytics system supported by Google Cloud, we believe that we can reach our expansion goals and continue innovating cinema viewing experiences."